PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (http://bmjopen.bmj.com/site/about/resources/checklist.pdf) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The risk of infection and contribution to transmission of SARS-CoV-2 in school staff - a systematic review
AUTHORS	Karki, Sudip Jung; Joachim, Alexandar; Heinsohn, Torben; Lange, Berit

VERSION 1 – REVIEW

REVIEWER	Surén, Pål Norwegian Institute of Public Health
REVIEW RETURNED	14-Jul-2021

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript. It covers an important public health topic. The systematic review appears to be well conducted, in keeping with the PRISMA guidelines. The supplementary tables provide a good overview of the studies that were included. I have the following comments: - Methods, page 6, line 12-13: Only studies that were published before January 29, 2021, are included. Thus, the analyses will (in most cases) not have captured the time period after the alpha variant became dominant. The emergence of new variants does not make older studies obsolete, but if this review is to have any impact on policy, it would be a great benefit to include information about the alpha variant. The emergence of the delta variant makes this even more important, given that it spreads even more rapidly than the alpha variant. - Results and Discussion: Throughout these sections, the authors draw the conclusions that secondary attack rates depend on the community incidence, such that the secondary attack rate increases when the community incidence increases. What is the basis of this interpretation? Isn't this more likely to be explained by multiple introductions of the virus into schools during periods of high community incidence, i.e., that more cases appear to be related even though they are not directly connected? One gets the impression that the SARS-CoV-2 virus somehow changes its capabilities and becomes more infectious when the community incidence is high. Is that what the authors meant? - Discussion, page 16, line 42-49: The statements about mass screening/testing are not supported by the evidence from the studies in the review. These studies have not investigated the

	sensitivity, specificity and cost-effectiveness of mass testing (as opposed to targeted tracing and testing after the discovery of an index case). Given the low secondary attack rates from infected children, a more plausible interpretation would be that mass testing in schools is probably not cost-effective – but again, this is not what the studies have investigated.

REVIEWER	Vermund, Sten Yale University School of Medicine, Yale School of Public Health
REVIEW RETURNED	03-Aug-2021

GENERAL COMMENTS	The paper is timely as the topic is of enormous importance to global educational policy in the COVID-19 era. The systematic review methods are sound and the conclusions match the papers that were systematically reviewed. My comments are not profound as I found the paper to be a very strong one. MAJOR COMMENTS 1. P.15, lines 12-17- Since universal testing was not done in all these studies (correct?), symptomatic cases are detected more reliably than asymptomatic cases. So attack rates should be for "symptomatic disease", correct? If it is a jumble of symptomatic and asymptomatic disease, we'll need some clarification of interpretation in the discussion. PP.15-16, lines 57-60 and 3-5 are the start of this discussion, but I think needs expansion. 2. P.16, line 35 and elsewhere- Critical to say "children seem less susceptible to symptomatic SARS-CoV-2 infection", adding the word "symptomatic". The papers do not guide us much on total infections as universal screening was not always deployed. MINOR COMMENTS 3. Please avoid acronyms in all table or figure titles/legends. Table one, define PICO, for example, and NPIs in Tables 3 and 4.. 4. Table 2 needs editing for consistency. For example, Source sometimes writes out entire journal name and sometimes does not (e.g., MMWR). First author sometimes complete name, sometimes just last time, sometimes last name and initials. Date collection period sometimes month is abbreviated, sometimes not, sometimes year omitted. The same is true for the supplemental tables; an edit for consistency would be helpful. 5. Table 3 and Table 4will be easier to read when published if lighter background colors are used. No need for background color in the two rows of headers (both tables) or first column (Table 4). 6. P.14, line 49- Would delete the word "recent" as all the studies meet a reasonable definition of recent. 7. P.15, line 8- When you say "rates are rather low" can you indicate your comparator? Compared to attack rates for what other environment? Or reword to get rid of the vague, "rather low".

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Pål Surén, Norwegian Institute of Public Health
Comments to the Author: Thank you for the opportunity to review this manuscript. It covers an important

public health topic. The systematic review appears to be well conducted, in keeping with the PRISMA guidelines. The supplementary tables provide a good overview of the studies that were included.

I have the following comments:
1. Methods, page 6, line 12-13: Only studies that were published before January 29, 2021, are included. Thus, the analyses will (in most cases) not have captured the time period after the alpha variant became dominant. The emergence of new variants does not make older studies obsolete, but if this review is to have any impact on policy, it would be a great benefit to include information about the alpha variant. The emergence of the delta variant makes this even more important, given that it spreads even more rapidly than the alpha variant.

Answer: While first results of this manuscript were originally used in the beginning of 2021 to provide policy advice, we very much agree that the inclusion of the studies after emergence of alpha, beta and delta variant would be able to provide greater impact on policy now. We have now more explicitly mentioned this as one of the limitations of this review (Page 15, Line 15-17).

2. Results and Discussion: Throughout these sections, the authors draw the conclusions that secondary attack rates depend on the community incidence, such that the secondary attack rate increases when the community incidence increases. What is the basis of this interpretation? Isn't this more likely to be explained by multiple introductions of the virus into schools during periods of high community incidence, i.e., that more cases appear to be related even though they are not directly connected? One gets the impression that the SARS-CoV-2 virus somehow changes its capabilities and becomes more infectious when the community incidence is high. Is that what the authors meant?

Answer: We agree that higher secondary attack rate during higher community incidence are more likely explained multiple introductions of the virus into schools during periods of high community incidence and therefore easier detection of these infections.
This is also a finding that becomes most apparent not in the individual study, but rather after stratification of study results by infection settings during their data collection as performed by us.
Due to heterogeneous nature of included studies, we stratified the findings based in community level incidence/infection dynamics during data collection and prevailing mitigation measures implemented during that period. In the result and discussion section, we then present the attack rate during low, medium and high community level incidence. Therefore, in these section we do not mean to say secondary attack rate depend on community level incidence. In addition, it was not our intention to create an impression to readers that the SARS-CoV-2 virus changes its capabilities and becomes more infectious when community incidence is high.
Now, taking your comment and impression into consideration, we have rephrased the 1st and 3rd paragraph of discussion (Page 14, Line 3-6, and 18-19) and some parts of findings (Page 12, line 3-4). We hope this will not misguide the reader now.

3. Discussion, page 16, line 42-49: The statements about mass screening/testing are not supported by the evidence from the studies in the review. These studies have not investigated the sensitivity, specificity and cost-effectiveness of mass testing (as opposed to targeted tracing and testing after the discovery of an index case). Given the low secondary attack rates from infected children, a more plausible interpretation would be that mass testing in schools is probably not cost-effective – but again, this is not what the studies have investigated.

Answer: We agree that mass screening/testing are not supported by the evidence from the studies that are included in the review. However, in our view the finding of relevant infection risks, attack rates and

secondary attack rates in the school setting is an argument for adhering to now well-laid out school hygiene measures and test strategies (please see also the German evidence based guideline on this. This is the link for the guideline https://www.bmbf.de/bmbf/shareddocs/kurzmeldungen/de/die-s3-leitlinie-als-handlungsempfehlung-fuer-schulen.html). School ministries and similar authority bodies have sometimes doubts the relevance of these measures based on the impression that no relevant transmission was taking place in the school setting. So, we believed it to be important to make this connection clear in the discussion. You are very correct however, in that we should not be overstating it.

Reviewer: 2
Dr. Sten Vermund, Yale University School of Medicine
Comments to the Author: The paper is timely as the topic is of enormous importance to global educational policy in the COVID-19 era. The systematic review methods are sound and the conclusions match the papers that were systematically reviewed. My comments are not profound as I found the paper to be a very strong one.
MAJOR COMMENTS
1. P.15, lines 12-17- Since universal testing was not done in all these studies (correct?), symptomatic cases are detected more reliably than asymptomatic cases. So attack rates should be for “symptomatic disease”, correct? If it is a jumble of symptomatic and asymptomatic disease, we’ll need some clarification of interpretation in the discussion. PP.15-16, lines 57-60 and 3-5 are the start of this discussion, but I think needs expansion.

Answer: We agree that universal testing was not done in all the included studies and we were not clear about whether they tested only symptomatic or reported case or both symptomatic and asymptomatic cases. Hence, we tried to present one scenario to highlight the influence of local arrangements and testing strategies on the results or comparison that we are making.
Based on this comment we rephrase the sentence in Page 15, line 2-3. We deleted “children are more likely to become asymptomatically ill” and added “Children experience a mild clinical course and fewer symptoms hence increase the chance of being untested or not reported.”

2. P.16, line 35 and elsewhere- Critical to say “children seem less susceptible to symptomatic SARS-CoV-2 infection”, adding the word “symptomatic”. The papers do not guide us much on total infections as universal screening was not always deployed.

Answer: Yes, we agree on your comment and the addition of word “symptomatic” would have made perfect sense in that particular line if we were sure about the testing strategies in included studies i.e. only if testing was done in symptomatic cases or reported cases. As you mentioned universal screening was not always deployed, likewise testing only symptomatic SARS-CoV-2 were not always the case. Therefore, we did not include the word symptomatic in on that particular line and elsewhere. We have now explained this limitation in more detail in the discussion (Page 15, Line 12-15).

MINOR COMMENTS
3. Please avoid acronyms in all table or figure titles/legends. Table one, define PICO, for example, and NPIs in Tables 3 and 4.

Answer: Full form of the above-mentioned term is written in the tables mentioned above in the comment.

4. Table 2 needs editing for consistency. For example, Source sometimes writes out entire journal name and sometimes does not (e.g., MMWR). First author sometimes complete name, sometimes just last time, sometimes last name and initials. Date collection period sometimes month is abbreviated, sometimes not,

sometimes year omitted. The same is true for the supplemental tables; an edit for consistency would be helpful.
Answer: Table 2 formatted accordingly.
⎫ Full form of MMWR i.e. Morbidity and Mortality Weekly Report is added.
⎫ First author name: All the names are changed to last name and initials.
⎫ Date: day followed by complete month name and year. We added the complete name of the months and year.

Supplementary Table 1
⎫ Full form of NCIRS i.e. National Centre for Immunisation Research and Surveillance is added.
Supplementary Table 2
⎫ Full form of NCIRS i.e. National Centre for Immunisation Research and Surveillance is added.
⎫ The full form is of AR i.e. Attack rate is added to the first row of Supplementary Table 2.
⎫ Date: day followed by complete month name and year. We added the complete name of the months and year
⎫ Full form of MMWR i.e. Morbidity and Mortality Weekly Report is added
Supplementary Table 3:
⎫ Date: day followed by complete month name and year. We added the complete name of the months and year
⎫ First author name: All the names are changed to last name and initials whenever applicable.
Supplementary Table 4:
⎫ Full form of NPIs i.e. Non-pharmaceutical Intervention is added
⎫ Full form of NCIRS i.e. National Centre for Immunisation Research and Surveillance is added.

5. Table 3 and Table 4 will be easier to read when published if lighter background colors are used. No need for background color in the two rows of headers (both tables) or first column (Table 4)

Answer: The background color in two rows of headers and first column of both tables 3 and 4 are removed.

6. P.14, line 49- Would delete the word “recent” as all the studies meet a reasonable definition of recent.

Answer: The word recent is deleted from the line (Page 13, Line 9).

7. P.15, line 8- When you say “rates are rather low” can you indicate your comparator? Compared to attack rates for what other environment? Or reword to get rid of the vague, “rather low”.

Answer: The comparator was the places with medium and high incidence of infection. We added the comparator in the sentence now (Page 14, line 4-5).

VERSION 2 – REVIEW

REVIEWER	Surén, Pål Norwegian Institute of Public Health
REVIEW RETURNED	21-Sep-2021

GENERAL COMMENTS	My comments from the previous review have been adequately addressed and responded to. I have no further comments.